



Fig. 6: The final assemblage enclosed in a cabinet (front panel)



Fig. 7: The final assemblage enclosed in a cabinet (internal wiring)



Fig. 8: The final assemblage enclosed in a cabinet (rear panel)

terminal of the battery using suitable arrangement. The value of this current limiter will depend on your requirement, so it is not shown in the circuit here.

DC voltage so generated is used to charge the battery. At the same time, terminal 9 of the relay comes into contact with terminal 6, which disconnects power supply to the oscillator circuit and deactivates the inverter circuit. Switch S1 should be open when the battery is fully charged, which will be indicated by lighting up of LED2.

When AC mains power is absent (power cut). When AC mains power is off, transformer X1 does not get 230V AC supply. Relay RL1 therefore does not energise. Terminals 7, 8

and 9 of the relay come into contact with terminals 1, 2 and 3, respectively. Terminal 9 is connected to the positive terminal of the 12V battery, which is extended to the oscillator circuit. Inverter circuit comes into action.

Transformer X2 is now a step-up transformer. AC voltage from 240V tap of transformer X2 is connected to terminal 1 of the relay. As terminals 7 and 8 are in contact with terminals 1 and 2 of the relay, 240V AC gets connected to output socket CON8. Neon lamp N1 is connected between terminal 2 and neutral. It glows when the UPS is on. The output is connected to 240V tapping because there will be a voltage drop when load is connected to the UPS.

Construction and testing

Combined actual-size, single-side PCB for the oscillator section and the battery-status section is shown in Fig. 4 and the component layout in Fig. 5.

If needed, the PCB can be cut into two portions along the dotted line and mounted separately. Rest of the circuit has been wired using connectors. The final assemblage of the mini off-line UPS is enclosed in a cabinet as shown in Fig. 6. The internal wiring is shown in Fig. 7 and the rear panel is shown in Fig. 8. All switches, indicators and terminals for connecting the battery and output socket are to be placed aesthetically on the front panel of the cabinet.

Fuse F1 (1A) is used to protect the device from any short circuits. All connections should be made very carefully. The load should not exceed 350VA. For troubleshooting, check voltages at various test points as listed in the table.

Caution. Please be careful as the circuit operates on 230V AC. ●



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